

Definition. The **length** of a smooth curve $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$ from $t = a$ to $t = b$ is given by

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt = \int_a^b \sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2} dt,$$

where we assume that the curve is traced exactly once on $[a, b]$.

Observe that if $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$, then $\mathbf{v}(t) = \mathbf{r}'(t) = \langle x'(t), y'(t), z'(t) \rangle$ and therefore we find that

$$L = \int_a^b \|\mathbf{v}(t)\| dt.$$

Example. A glider is soaring upward along the helix $\mathbf{r}(t) = \langle \cos(t), \sin(t), t \rangle$. How long is the glider's path from $t = 0$ to $t = 2\pi$?

A common question one might ask is “Suppose we only start looking at the glider after t_0 seconds have passed. How far will it travel once we get to t seconds?” Here, we are asking for a function $s(t)$ for which, say, $s(t)$ tells us how far the glider has traveled in the $t - t_0$ seconds we've looked at it:

$$s(t) = \int_{t_0}^t \|\mathbf{v}(\tau)\| d\tau$$

Example. Reconsider the curve $\mathbf{r}(t) = \langle \cos(t), \sin(t), t \rangle$ over $0 \leq t \leq 2\pi$. Find the arclength parameter along the curve starting at $t_0 = 0$ and use this to find the length of the curve.

Example. Consider $\mathbf{r}(t) = \langle 7 + 5t, 2 + 7t, 1 - 3t \rangle$ over $-1 \leq t \leq 0$. Find the arclength parameter along the curve starting at $t_0 = 0$ and use this to find the length of the curve.

Unit Tangent Vector

Given a position vector $\mathbf{r}(t)$, we know that its velocity $\mathbf{v}(t) = \mathbf{r}'(t)$ is tangent to $\mathbf{r}(t)$. But there is no reason to believe that this vector is *unit*. Thus, we define

$$\mathbf{T}(t) = \frac{\mathbf{v}(t)}{\|\mathbf{v}(t)\|}$$

to be the directional unit vector pointing in the direction of $\mathbf{v}(t)$. This is called the **unit tangent vector**.

Example. Find the unit tangent vector of the curve $\mathbf{r}(t) = \langle 0, \cos^3(t), \sin^3(t) \rangle$.

Speed on a Smooth Curve

Since

$$s(t) = \int_{t_0}^t \|\mathbf{v}(\tau)\| d\tau,$$

we can differentiate both sides to see that

$$\frac{ds}{dt} = \|\mathbf{v}(t)\|.$$

Notice that this makes sense: s records the ‘distance traveled’, so $\frac{ds}{dt}$ is the change in distance over the change in time: this is *exactly* what speed is!

The last thing we look at is the following: We already know that

$$\frac{d\mathbf{r}}{dt} = \mathbf{v}(t).$$

That is, the change in position *with respect to time* is the velocity vector. What if we change our point of reference? For example, what if we want to know the change in position **with respect to arclength**:

$$\frac{d\mathbf{r}}{ds}.$$

That is, what vector records how quickly position changes as we move along the curve traced out by $\mathbf{r}$? Let’s find out:

$$\frac{d\mathbf{r}}{ds} =$$

That is, the change in position *with respect to arclength* is that unit tangent vector in the direction of the velocity vector. We will need this fact later on in Sections 12.4 and 12.5.